

MANU PRATHAP

Frontend Engineer | Full-Stack Developer | AI-Integrated Web Applications

■ +91 70254 05061 | ✉️ manuprathapkarimbil@gmail.com | ■ linkedin.com/in/manuuprathaap | ■ Kochi, India

PROFESSIONAL SUMMARY

Results-driven **Frontend Engineer with 3+ years** of experience architecting scalable, high-performance web applications using **Angular (8–18+), React.js, and TypeScript**. Proven track record of **reducing load times by 40%+**, optimizing bundle sizes, and delivering production-grade SPAs with SSR and PWA capabilities. Expanding expertise in **AI/LLM integration** and full-stack development (Node.js, Python, Firebase). Passionate about clean architecture, measurable performance gains, and building user-centric digital products.

TECHNICAL SKILLS

Frontend Expert:	Angular 8–18, React.js, Vue.js, TypeScript, JavaScript (ES6+), HTML5, CSS3/SCSS
State & Performance:	RxJS, NgRx, Signals, Lazy Loading, OnPush, SSR, PWA, Code Splitting, Virtual Scrolling
UI & Libraries:	Angular Material, PrimeNG, Bootstrap, Chart.js, Three.js, Responsive Design
Backend & APIs:	Node.js, Express.js, Python, Django, REST APIs, GraphQL, WebSockets, HTTP Interceptors
Databases & Cloud:	Firebase (Auth, Firestore, Hosting), MongoDB, MySQL, AWS (EC2, S3), Docker
AI & Emerging:	LLM Integration, Prompt Engineering, RAG Pipelines, AI Agents, CI/CD, Git, Agile/Scrum

PROFESSIONAL EXPERIENCE

Frontend Engineer

Dec 2023 – Dec 2025

ElkanIO Research Labs, Kochi, India

- Architected and deployed **3 production Angular SPAs** with SSR and PWA capabilities, **improving Lighthouse performance scores from 62 to 94** and boosting SEO rankings for client platforms.
- Reduced initial page load time by **40%** and bundle size by **35%** through strategic implementation of lazy loading, OnPush change detection, and modular architecture across all projects.
- Designed centralized REST API integration layer with Angular HttpClient interceptors for token-based auth, error handling, and request/response transformation — **cutting API-related bugs by 60%** and standardizing data flow.
- Built scalable state management solutions using **NgRx and RxJS**, establishing reusable component patterns that **accelerated feature delivery by 25%** across the development team.
- Mentored 2 junior developers on Angular best practices, code review standards, and performance optimization — improving team code quality scores and reducing review cycles.

Frontend Developer

Jan 2023 – Dec 2023

Exouzia, India

- Developed reusable UI components and feature modules using **Angular, TypeScript, and JavaScript**, focusing on UX consistency and component-driven architecture for scalable delivery.
- Integrated RESTful APIs and third-party plugins to enable seamless inter-application communication, supporting real-time data synchronization and enhanced feature sets.
- Maintained clean, maintainable codebases with consistent attention to performance optimization and cross-browser compatibility across all deliverables.

KEY PROJECTS

iHub-24 — Restaurant POS System

Angular 18, Firebase, Firestore, REST API

- Built a real-time POS system with **zero-refresh order updates** using Firebase Firestore real-time listeners. Delivered role-specific dashboards (cashier, kitchen, admin) with live order tracking. **Sole developer** handling full SDLC from architecture to deployment on Firebase Hosting.

TripXOXO — Digital Travel Platform

Angular 16, RxJS, REST API, Jira

■ Engineered a dynamic drag-and-drop form builder with custom reusable pipes and directives. Implemented REST API consumption with HttpClient interceptors for auth and centralized error handling. Practiced Agile sprints with AI-assisted planning.

BookOTrip — Online Travel Booking Platform

Angular 16, REST API, MySQL

■ Developed admin and user modules for domestic/international flight and hotel booking with fully responsive UI. Implemented real-time booking workflows with reactive state management and REST API integration.

Stamp the Passport — Visa Application Portal

Angular 16, Signzy API, Reactive Forms

■ Designed dynamic country-specific visa questionnaires with document upload workflows. Integrated Signzy APIs for passport verification with real-time identity validation. Scalable form architecture with client-side and server-side validation, **reducing manual review time by 50%**.

GeoPerform — Vessel Performance Management

Angular 15, Role-Based Access Control

■ Developed admin and RBAC modules for vessel performance monitoring, enabling secure multi-role user management and real-time data visualization dashboards.

EDUCATION & CERTIFICATIONS

B.Tech — Computer Science Engineering

2019 – 2022

APJ Abdul Kalam Technological University | Royal College of Engineering

Diploma — Computer Engineering

2016 – 2019

Govt. Polytechnic College Kunnankulam, Kerala

Generative AI & Agentic AI — In Progress (2026)

LLM fundamentals, prompt engineering, RAG pipelines, AI agent development